MD. MIZANUR RAHMAN

Software Engineer | React, TypeScript, Node.js, Rust

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TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Java, Python, Rust, C#, C++, Kotlin, SQL
Frontend	React, Next.js, Tauri, three.js
Backend	Node.js, Express, FastAPI, React Native, ASP.NET Core, REST APIs
Databases	PostgreSQL, MySQL, SQLite, MongoDB (NoSQL)
DevOps and Testing	Docker, Kubernetes, AWS, Git, NPM, Linux, unit testing with pytest
Security	ISO 27001, ISMS, risk assessment, OAuth 2.0, JWT

EXPERIENCE

Samsung R&D Institute Bangladesh — Dhaka, Bangladesh
Software Engineer — *Mar 2024 – Sep 2026*

Device Manager | *Tauri, React, TypeScript, Rust, Node.js, Express*

- Delivered a cross-platform desktop app for managing local and remote Tizen devices, writing **80% of the codebase**.
- Built the entire **React and TypeScript** UI and the **Rust** backend that fetches, filters, and updates device data.
- Built an **Express REST API** that connects and disconnects devices over IP, retrieves and filters device logs, and scans the subnet to discover devices.

TMPortal | *React, REST APIs*

- Replaced the default layout components with a custom **sidebar and navigation menu** built in React.
- Built **reusable tables, forms, and dialogs** shared across every portal page, keeping the interface consistent.
- Integrated pages with backend **REST APIs**, handling loading, empty, and error states on each view.

Samsung Ring Plugin | *Android, iOS*

- Unified all Android byte parsers into one **generic parser** with full backward compatibility.
- Designed **byte protocols** for new health-sensing modules and shipped those modules end to end.
- Resolved user-reported bugs by correlating ring binary data with plugin and Samsung Health logs.

Tizen Doctor CLI | *JavaScript, Node.js*

- Automated environment checks for .NET, Web, and Native Tizen apps in Visual Studio and VS Code on **Linux, macOS, and Windows**.
- Built diagnostics that detect and fix common SDK and build-tool problems and tell developers how to resolve the rest.
- Wrote the tool's complete **official documentation**.

Visual Studio Extension for Tizen | *C#, .NET*

- Enabled **debugging and app installation on RISC-V** Tizen devices directly from Visual Studio.
- Shipped features and fixes for the extension and Baseline SDK tools (Package, Device, Emulator Manager) with each **Tizen release**.
- Wrote the **official documentation** for new features and for older features that had none.

Makebell Limited — Hong Kong SAR
Security Engineer and ISO 27001 Consultant — *Jun 2026 – Aug 2026*

- Built a custom **MDR** (managed detection and response) platform end to end, with a **Rust** agent that monitors company devices and streams security telemetry to a central dashboard for alerting.
- Completed the **ISO 27001 Stage 2 certification audit** by building the ISMS end to end, including the risk register, risk treatment plan, and Statement of Applicability.

PROJECTS

PrintForge, 3D Model Marketplace with AI Image-to-3D | *Next.js, TypeScript, PostgreSQL, three.js, NextAuth* | *GitHub*

- Built a full-stack marketplace on the **Next.js App Router** and **PostgreSQL** with an interactive **three.js** glTF viewer, search, filters, server-side pagination, and optimistic likes.
- Integrated **AI image-to-3D generation** through the Hunyuan3D-2 Gradio API, streaming job results over **Server-Sent Events** and enforcing timeouts and one job per user on a shared GPU quota.
- Built a **50 MB upload pipeline** that streams to disk, validates glTF headers and byte counts, and saves atomically with rollback, so a failed upload never leaves a broken record.
- Secured the app with **NextAuth** (bcrypt, JWT), server-side ownership checks, open-redirect protection, zod validation, and parameterized SQL queries.

NextStep: AI-powered application tracker | *React Native (Expo), TypeScript, Firebase, Supabase (Postgres + pgvector), Gemini API* | *GitHub*

- Built a cross-platform **Expo Router app** to track job, university and professor applications. It syncs live through **Firebase Realtime Database**, accepts edits offline, and has a deadline calendar, scheduled reminders, full-screen Android alarms and follow-up nudges.
- Built a **RAG chat** over users' saved job circulars and notes. It uses **section-aware chunking, Gemini embeddings, and hybrid pgvecto** and full-text search merged with reciprocal rank fusion, and every answer links to its sources.
- Integrated **LLM extraction** that turns a pasted circular into form data as schema-constrained JSON, and checks each deadline, link and email against the source text. Cut chat latency from 19 s to 2.3 s by benchmarking model and reasoning settings.
- Secured the AI backend with a Supabase Edge Function that verifies **Firebase JWTs** against Google's public keys, deny-by-default Row Level Security, atomic per-user daily quotas and prompt-injection defenses. Re-embeds only changed chunks, detected with SHA-256 hashes.

FastAPI Blog Platform | *Python, FastAPI, PostgreSQL, JavaScript, Bootstrap, AWS S3, Docker* | *GitHub*

- Built an async **FastAPI REST API** with a Jinja2, Bootstrap 5, and vanilla JavaScript frontend that updates votes live and paginates feeds without page reloads.
- Secured accounts with **OAuth 2.0 and JWT**, Argon2 hashing, and owner-only writes, storing single-use reset tokens as SHA-256 hashes that expire in 60 minutes.
- Designed a **PostgreSQL** schema that enforces one vote per user, migrated from SQLite with **Alembic**.
- Moved blocking S3 uploads to a threadpool to keep the event loop free, isolated each **pytest** test in a rolled-back transaction, and shipped a multi-stage **Docker** image.

Kubernetes App Deployment | *Next.js, MongoDB, Docker, Kubernetes, AWS* | *GitHub*

- Containerized a Next.js and MongoDB app with **Docker Compose** and deployed each component to **Kubernetes** as a Deployment behind a ClusterIP Service.
- Routed all external traffic through a single **NGINX Ingress**, kept config in ConfigMaps and credentials in Secrets, and ran the stack on minikube and **AWS**.

Chronic Kidney Disease Detection | *Python, Machine Learning, Streamlit* | *GitHub*

- Trained and compared several classifiers on clinical data such as blood pressure, blood cell counts, and sugar level, and deployed the best model as a **Streamlit** web app.

Brain Tumor Segmentation | *Undergraduate Thesis, Python, Deep Learning*

- Built a hybrid **3D MRI segmentation model** based on a modified U-Net with dynamic convolution and attention, including the data pipeline, training loop, and evaluation.

EDUCATION

Khulna University of Engineering & Technology (KUET) — Khulna, Bangladesh
B.Sc. in Computer Science and Engineering, CGPA 3.43 — *2019 – 2024*

ACHIEVEMENTS AND ACTIVITIES

- Passed the **Samsung SWC Professional** exam (certificate).
- Served as Assistant General Secretary of Bit2Byte and as a member of SGIPC, the KUET programming club.